

# Evidence-Based Pedagogy for AI Teaching Assistants: A Technical Reference

An AI Teaching Assistant built on cognitive science principles can dramatically improve learning outcomes. This reference synthesizes meta-analyses, peer-reviewed research, and effect size data across four domains—the cognitive engine, instructional blueprints, classroom mechanics, and lecture techniques—providing specific algorithms, timing protocols, and structured data ready for system prompt implementation.

## MODULE 1: The Cognitive Science of Learning (The "Engine")

### 1.1 Cognitive Load Theory

#### Working Memory Constraints

| Parameter                   | Value       | Source                |
|-----------------------------|-------------|-----------------------|
| Classic capacity            | 7 ± 2 items | Miller (1956)         |
| Modern estimate             | 3-4 chunks  | Cowan (2001)          |
| Duration without rehearsal  | ~20 seconds | Sweller (2011)        |
| Concurrent processing limit | 2-4 chunks  | Paas & Sweller (2012) |

#### Three Load Types

TOTAL\_LOAD = intrinsic + extraneous + germane

WHERE:

intrinsic = content complexity (fixed by material)

extraneous = poor design overhead (TARGET FOR REDUCTION)

germane = schema construction effort (maximize this)

### 5 Techniques to Reduce Extraneous Load

| Technique              | Protocol                                                         | Effect Size |
|------------------------|------------------------------------------------------------------|-------------|
| 1. Worked Examples     | Present complete solutions before practice; fade steps gradually | d = 0.57    |
| 2. Spatial Integration | Embed labels INTO diagrams—never separate text from graphics     | d = 0.79    |
| 3. Modality Effect     | Use audio narration + visuals (not text + visuals)               | d = 0.72    |
| 4. Coherence Principle | Remove decorative images, background music, extra words          | d = 0.70    |
| 5. Segmenting          | Break content into learner-paced 5-7 minute chunks               | d = 0.70    |

Hattie Effect Size: Cognitive load interventions = **d = 0.57** (above 0.40 hinge point)

1.2 Retrieval Practice & Spaced Repetition

The Testing Effect (Meta-Analysis Data)

| Study                 | Effect Size                  | Finding                                  |
|-----------------------|------------------------------|------------------------------------------|
| Rowland (2014)        | $g = 0.50$                   | Retrieval vs. reading (159 effect sizes) |
| Adesope et al. (2017) | $g = 0.61$                   | Retrieval vs. all other strategies       |
| With feedback         | $d = 0.73$                   | Testing + corrective feedback            |
| Without feedback      | $d = 0.39$                   | Testing alone                            |
| Hattie                | <b><math>d = 0.54</math></b> | Practice testing                         |

Ebbinghaus Forgetting Curve

| Time Post-Learning | Retention    |
|--------------------|--------------|
| End of session     | ~75%         |
| 20 minutes         | ~58%         |
| 1 hour             | ~44%         |
| 1 day              | ~ <b>33%</b> |
| 1 week             | ~25%         |
| 1 month            | ~20%         |

Optimal Spacing Intervals

*Pimsleur Schedule (1967):*

|                        |                       |
|------------------------|-----------------------|
| Interval 1: 5 seconds  | Interval 7: 1 day     |
| Interval 2: 25 seconds | Interval 8: 5 days    |
| Interval 3: 2 minutes  | Interval 9: 25 days   |
| Interval 4: 10 minutes | Interval 10: 4 months |
| Interval 5: 1 hour     | Interval 11: 2 years  |
| Interval 6: 5 hours    |                       |

*SM-2 Algorithm (SuperMemo):*

|        |
|--------|
| python |
|--------|

```
def calculate_next_interval(n, EF, quality):
    """
    n = review number (1, 2, 3...)
    EF = Ease Factor (starts 2.5, min 1.3)
    quality = response quality (0-5)
    """
    # Update Ease Factor
    EF_new = EF + (0.1 - (5 - quality) * (0.08 + (5 - quality) * 0.02))
    EF_new = max(1.3, EF_new)

    # Calculate interval
    if n == 1: return 1 # day
    if n == 2: return 6 # days
    return round(previous_interval * EF_new)
```

### Quality Scale:

- 5 = Perfect, instant recall
- 4 = Correct after hesitation
- 3 = Correct with difficulty
- 2 = Incorrect, but seemed easy
- 1 = Incorrect, remembered upon seeing
- 0 = Complete blackout

### Make It Stick Core Findings

- Retrieval practice >> re-reading (even unsuccessful attempts help) (Danielschristian)
- Spacing >> cramming (Danielschristian)
- Interleaving (mixing topics) >> blocked practice (Danielschristian)
- **Desirable difficulties** strengthen long-term retention (Amazon)
- Students incorrectly prefer ineffective strategies (fluency illusion) (Retrieval Practice)

## 1.3 Dual Coding Theory & Multimedia Learning

### Mayer's Principles with Effect Sizes

| Principle           | Description                                      | Effect Size |
|---------------------|--------------------------------------------------|-------------|
| Multimedia          | Words + pictures > words alone                   | d = 1.67    |
| Temporal Contiguity | Present narration and graphics simultaneously    | d = 1.30    |
| Redundancy          | Don't duplicate narration with on-screen text    | d = 0.87    |
| Spatial Contiguity  | Place text near corresponding graphics           | d = 0.79    |
| Personalization     | Use conversational "you" language                | d = 0.79    |
| Modality            | Audio narration > on-screen text (with graphics) | d = 0.72    |
| Coherence           | Eliminate extraneous content                     | d = 0.70    |
| Segmenting          | Learner-controlled pacing                        | d = 0.70    |
| Signaling           | Highlight key information                        | d = 0.46    |

Visual-Verbal Integration Rules

RULE 1: CONTIGUITY

IF text explains graphic THEN embed text INTO graphic

RULE 2: MODALITY

IF visual is complex THEN use audio narration (not on-screen text)

RULE 3: REDUNDANCY

IF narration exists THEN DO NOT show identical text

RULE 4: EXPERTISE REVERSAL

IF learner is expert THEN reduce scaffolding (it becomes load)

IF learner is novice THEN maximize scaffolding

1.4 Metacognition

EEF Toolkit Data

| Metric            | Value                               |
|-------------------|-------------------------------------|
| Impact            | +7 to +8 months additional progress |
| Cost              | Very low (£)                        |
| Evidence strength | High (355 studies)                  |

Hattie Effect Sizes

| Strategy                       | Effect Size     |
|--------------------------------|-----------------|
| Metacognitive strategies       | <b>d = 0.69</b> |
| Self-verbalization/questioning | d = 0.64        |
| Elaboration and organization   | d = 0.75        |

## AI Prompt Library: Metacognitive Questions

### *Planning Phase:*

- "What is this task actually asking you to do?"
- "What do you already know about [topic]?"
- "What strategy might work best here?"
- "Have you solved something similar before?"

### *Monitoring Phase:*

- "Am I on the right track?"
- "Does this answer make sense so far?"
- "What should I do if I get stuck?"
- "How confident am I right now (1-5)?"

### *Evaluating Phase:*

- "Did my strategy work? Why or why not?"
- "What would I do differently next time?"
- "Can I explain this to someone else?"
- "What was the hardest part and how did I overcome it?"

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## MODULE 2: Instructional Design Frameworks (The "Blueprint")

### 2.1 Gagné's Nine Events of Instruction

#### Algorithm with AI Activities

## EVENT 1: GAIN ATTENTION (2-5 min)

Purpose: Activate orienting response

AI Activities:

1. Present surprising statistic related to topic
2. Pose "mystery question" lesson will answer
3. Show thought-provoking image with reflection prompt

AI Prompt: "Did you know that [counterintuitive fact]?  
What's your initial reaction?"



## EVENT 2: STATE OBJECTIVES (1-2 min)

Purpose: Set expectations and relevance

AI Activities:

1. Display "By end of lesson, you will be able to..."
2. Connect objectives to learner's stated goals
3. Show visual roadmap of lesson progression

Format: [Action Verb] + [Specific Content] + [Context]



## EVENT 3: STIMULATE PRIOR KNOWLEDGE (3-5 min)

Purpose: Create cognitive hooks for new information

AI Activities:

1. Quick retrieval quiz on prerequisites
2. "Explain what you know about [related concept]"
3. Concept map showing prior → new connections

IF misconception\_detected: provide corrective feedback



## EVENT 4: PRESENT CONTENT (10-20 min, chunked)

Purpose: Deliver new information in digestible segments

AI Activities:

1. Multi-modal presentation (text, audio, visual)
2. Chunk into 5-7 minute segments with transitions
3. Provide examples AND non-examples for each concept

AFTER each chunk: insert comprehension check



EVENT 5: PROVIDE GUIDANCE (interspersed)

Purpose: Support encoding and schema construction

AI Activities:

1. Mnemonic devices for key concepts
2. Graphic organizers showing relationships
3. Model thinking: "Let me show how to approach this..."

Gradually FADE scaffolding as competence increases



EVENT 6: ELICIT PRACTICE (10-15 min distributed)

Purpose: Apply learning; confirm understanding

AI Activities:

1. Low-stakes practice with graduated difficulty
2. Application scenarios requiring knowledge transfer
3. Peer discussion of solutions

Difficulty = learner.current\_mastery + 0.15 (ZPD)



EVENT 7: PROVIDE FEEDBACK (immediate)

Purpose: Confirm correct performance or guide correction

Hattie Effect Size:  $d = 0.70$

AI Activities:

1. Immediate, specific feedback on attempts

- |                                                            |  |
|------------------------------------------------------------|--|
| 2. Explain WHY answer is correct/incorrect                 |  |
| 3. Scaffold toward correct understanding with hints        |  |
| IF correct: "Correct! This works because [explanation]"    |  |
| IF incorrect: "Your thinking shows [X]. The issue is [Y]." |  |



|                                                             |
|-------------------------------------------------------------|
| EVENT 8: ASSESS PERFORMANCE (5-10 min)                      |
| Purpose: Verify learning objectives achieved                |
| AI Activities:                                              |
| 1. Varied assessments aligned to objectives                 |
| 2. Multiple demonstration formats (written, verbal, visual) |
| 3. Adaptive difficulty based on performance                 |
| Mastery threshold: typically 80%+                           |



|                                                           |
|-----------------------------------------------------------|
| EVENT 9: ENHANCE RETENTION & TRANSFER (end + spaced)      |
| Purpose: Long-term storage; application in new contexts   |
| AI Activities:                                            |
| 1. Connect to real-world applications relevant to learner |
| 2. Schedule spaced retrieval at optimal intervals         |
| 3. "Create your own example" / "Teach this to a friend"   |
| Spaced review: Day 1 → Day 3 → Day 7 → Day 14+            |

## 2.2 Understanding by Design (Backwards Design)

### WHERE TO Framework Decision Tree

python



```
def backwards_design(topic):
```

```
# =====  
# STAGE 1: IDENTIFY DESIRED RESULTS (Start with the End)  
# =====  
  
# Step 1.1: Enduring Understandings  
enduring = ask("What should students remember 5 years from now?")  
validate(enduring, criteria=[  
    "Is it a transferable big idea?",  
    "Does it have lasting value beyond classroom?",  
    "Is it central to the discipline?"  
)  
  
# Step 1.2: Essential Questions  
questions = ask("What open-ended questions will drive inquiry?")  
validate(questions, criteria=[  
    "Is it arguable/debatable?",  
    "Does it recur across the discipline?",  
    "Does it provoke genuine inquiry?"  
)  
  
# Step 1.3: Knowledge & Skills  
categorize_by_priority([  
    "Enduring understandings" → HIGHEST,  
    "Important to know and do" → MEDIUM,  
    "Worth being familiar with" → LOWEST  
)  
  
# =====  
# STAGE 2: DETERMINE ACCEPTABLE EVIDENCE (Think Like Assessor)  
# =====  
  
# Use GRASPS for authentic performance tasks:  
performance_task = {  
    "G": "Goal - What must be accomplished?",  
    "R": "Role - What role does student assume?",  
    "A": "Audience - Who is the target audience?",  
    "S": "Situation - What is the real-world context?",  
    "P": "Product - What will be created/performed?",  
    "S": "Standards - What criteria define success?"  
}
```

```
# Align each assessment ↔ specific learning objective
```

```
for objective in learning_objectives:
```

```
    assessment = design_assessment(objective.cognitive_level)
```

```
    verify_alignment(objective, assessment)
```

```
#
```

```
# STAGE 3: PLAN LEARNING EXPERIENCES (Design Instruction)
```

```
#
```

```
# Apply WHERETO:
```

```
return {
```

```
    "W": "Where going? Why? What expected?",
```

```
    "H": "Hook students; Hold attention",
```

```
    "E": "Equip with experiences, tools, knowledge",
```

```
    "R": "Rethink, reflect, revise understanding",
```

```
    "E": "Evaluate progress; self-assess",
```

```
    "T": "Tailor to individual needs",
```

```
    "O": "Organize for optimal learning"
```

```
}
```

## 2.3 Universal Design for Learning (UDL) Checklist

### Principle 1: Multiple Means of ENGAGEMENT (The "Why")

| Checkpoint               | Implementation                                     | ✓                        |
|--------------------------|----------------------------------------------------|--------------------------|
| Offer choices            | Let learners select content pathways, tools        | <input type="checkbox"/> |
| Optimize relevance       | Connect to learner's interests/goals               | <input type="checkbox"/> |
| Minimize threats         | Normalize mistakes; reduce anxiety                 | <input type="checkbox"/> |
| Vary challenge           | Adjust difficulty to Zone of Proximal Development  | <input type="checkbox"/> |
| Foster collaboration     | Enable peer interaction options                    | <input type="checkbox"/> |
| Provide mastery feedback | Focus on effort and strategy, not just correctness | <input type="checkbox"/> |
| Develop self-regulation  | Teach coping strategies; prompt reflection         | <input type="checkbox"/> |

### Principle 2: Multiple Means of REPRESENTATION (The "What")

| Checkpoint                   | Implementation                              | ✓                        |
|------------------------------|---------------------------------------------|--------------------------|
| Customize display            | Adjustable text size, font, contrast        | <input type="checkbox"/> |
| Offer audio alternatives     | Text-to-speech, narration                   | <input type="checkbox"/> |
| Offer visual alternatives    | Captions, transcripts, images               | <input type="checkbox"/> |
| Clarify vocabulary           | Glossary access, definitions on demand      | <input type="checkbox"/> |
| Illustrate via multimedia    | Use multiple media to explain concepts      | <input type="checkbox"/> |
| Activate prior knowledge     | Pre-teach prerequisites                     | <input type="checkbox"/> |
| Highlight patterns           | Use advance organizers, emphasize big ideas | <input type="checkbox"/> |
| Guide information processing | Provide graphic organizers                  | <input type="checkbox"/> |

**Principle 3: Multiple Means of ACTION & EXPRESSION (The "How")**

| Checkpoint                  | Implementation                             | ✓                        |
|-----------------------------|--------------------------------------------|--------------------------|
| Vary response methods       | Accept typing, voice, drawing, clicking    | <input type="checkbox"/> |
| Support assistive tech      | Compatible with screen readers, etc.       | <input type="checkbox"/> |
| Use multiple media          | Accept text, audio, video, diagrams        | <input type="checkbox"/> |
| Build fluencies             | Scaffold with templates, sentence starters | <input type="checkbox"/> |
| Guide goal-setting          | Break tasks into smaller steps             | <input type="checkbox"/> |
| Support planning            | Provide checklists, progress trackers      | <input type="checkbox"/> |
| Enhance progress monitoring | Prompt self-monitoring questions           | <input type="checkbox"/> |

**Neurodiverse Accommodations Quick Reference**

| Learner Profile     | Key Accommodations                                                                                                                                                                                                                                                                                         |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADHD                | Pomodoro (25+5 min), visual timers, transition warnings (2 min countdown reduces disruptions 72%), chunked assignments with checkpoints (↑ completion 57%), movement breaks                                                                                                                                |
| Dyslexia            | Text-to-speech, dyslexia-friendly fonts (OpenDyslexic), cream backgrounds, increased line spacing, accept oral responses, video + interactive content (retention ↑ from 54% to 82%)                                                                                                                        |
| Auditory Processing | Pair ALL verbal info with visual/written cues, (LD OnLine) captions on video, slow speech rate, (Heartwisesupport) repeat key info, (New Jersey Education Associat...) written copies of instructions, (Brain Balance Centers) extra processing time (10+ sec critical) (New Jersey Education Associat...) |
| Visual Processing   | Audio descriptions, high contrast, increased font sizes, tactile alternatives, verbal description of diagrams                                                                                                                                                                                              |

# MODULE 3: Classroom Organization & Logistics (The "Mechanics")

## 3.1 The Flipped Classroom

### Optimal Ratios

| Component                     | Ratio    | Time (3-credit course) |
|-------------------------------|----------|------------------------|
| Pre-class preparation         | 0.75-1.0 | <3 hours/week          |
| In-class active learning      | 1.0      | 3 hours/week           |
| Total in-class : out-of-class | 1:2      | —                      |

### Video Length Guidelines

| Duration     | Recommendation                            |
|--------------|-------------------------------------------|
| 6 minutes    | Maximum for K-12 (avoid disengagement)    |
| 8-12 minutes | Ideal length (research consensus)         |
| <20 minutes  | Total pre-class activity limit per lesson |

### Effect Sizes

| Influence            | Effect Size | Source       |
|----------------------|-------------|--------------|
| Flipped classrooms   | d = 0.58    | Hattie/Kapur |
| Classroom discussion | d = 0.82    | Hattie       |
| Cooperative learning | d = 0.40    | Hattie       |
| EEF MathsFlip trial  | +1 month    | EEF Toolkit  |

### Failure Prevention Matrix

| Failure Mode                | Prevention                                                  |
|-----------------------------|-------------------------------------------------------------|
| Students skip pre-work      | Embed quizzes in videos; accountability measures            |
| Pre-class overload          | Limit to 20 min max; 6-min video chunks                     |
| Reduced face-time           | Never cut in-class time; use for active learning            |
| No structure for activities | Plan in-class activities BEFORE selecting pre-class content |

## 3.2 Active Learning Structures

### THINK-PAIR-SHARE (Lyman, 1981)

| Spec        | Value                                  |
|-------------|----------------------------------------|
| Group size  | <b>2 (pairs)</b>                       |
| Total time  | <b>5-10 minutes</b>                    |
| Effect size | $d = 0.40-0.55$ (cooperative learning) |

**PROTOCOL:**

|                                                       |  |
|-------------------------------------------------------|--|
| THINK (1-3 min)                                       |  |
| → Pose higher-order question (not fact-based)         |  |
| → "Take 1-2 minutes to think and jot your response"   |  |
| PAIR (2-5 min)                                        |  |
| → "Turn to partner. A speaks first for 1 min, switch" |  |
| → Circulate and listen to discussions                 |  |
| SHARE (2-5 min)                                       |  |
| → "Bring it back in 3, 2, 1"                          |  |
| → Cold call: "Emily, what did Sarah tell you?"        |  |

**JIGSAW METHOD (Aronson, 1971)**

| Spec               | Value                                                   |
|--------------------|---------------------------------------------------------|
| Home group size    | <b>4-6 students</b>                                     |
| Expert group size  | <b>3-6 students</b>                                     |
| Total time         | <b>45-60 minutes</b>                                    |
| Hattie Effect Size | $d = 1.20$ (highest tier) <span>Visible Learning</span> |

**PROTOCOL:**

- Step 1: Form home groups (2-3 min) → Heterogeneous groups of 4-6
- Step 2: Assign segments (2 min) → Each member gets different content
- Step 3: Individual study (5-10 min) → Students read assigned segment
- Step 4: Expert groups form (2-3 min) → Same-segment students meet
- Step 5: Expert discussion (10-15 min) → Clarify, become "experts"
- Step 6: Return home (2 min) → Experts return to original groups
- Step 7: Peer teaching (15-20 min) → Each expert teaches their segment
- Step 8: Assessment (10-15 min) → Group task requiring ALL segments

| Use When                                         | Avoid When                         |
|--------------------------------------------------|------------------------------------|
| Content divides into equal, self-contained parts | Content is highly sequential       |
| Interdependence desired                          | Students lack collaborative skills |
| Multiple perspectives needed                     | Time severely limited (<30 min)    |

## PEER INSTRUCTION (Mazur, Harvard)

| Spec                 | Value                                                              |
|----------------------|--------------------------------------------------------------------|
| Class size           | <b>30-500+ students</b>                                            |
| ConcepTests per hour | <b>6-8</b>                                                         |
| Learning gain        | Traditional $g=0.23$ → Peer Instruction <b><math>g=0.48</math></b> |

### PROTOCOL (8-12 min per question):

|                                                                                                                                                                                       |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 1. Present concept (3-5 min)                                                                                                                                                          |  |
| 2. Pose ConcepTest MCQ (1 min)                                                                                                                                                        |  |
| 3. Individual think - NO discussion (1-2 min)                                                                                                                                         |  |
| 4. First vote (30 sec)                                                                                                                                                                |  |
| 5. Check response distribution                                                                                                                                                        |  |
| <div> <div>&lt;30% correct → Re-teach, NO peer discussion</div> <div>30-70% correct → PROCEED to peer discussion</div> <div>&gt;70% correct → Brief explanation, move on</div> </div> |  |
| 6. Peer discussion (2-3 min)                                                                                                                                                          |  |
| 7. Second vote (30 sec)                                                                                                                                                               |  |
| 8. Instructor explains (2-3 min)                                                                                                                                                      |  |

## FISHBOWL DISCUSSION

| Spec              | Value                        |
|-------------------|------------------------------|
| Inner circle      | <b>3-6 students</b> (active) |
| Outer circle      | <b>Remaining</b> (observers) |
| Time per rotation | <b>15-20 minutes</b>         |
| Total time        | <b>30-60 minutes</b>         |

Best for: Controversial topics, building multiple perspectives, practicing listening/speaking

## GALLERY WALK

| Spec             | Value                      |
|------------------|----------------------------|
| Group size       | <b>3-5 per group</b>       |
| Stations         | <b>4-8</b>                 |
| Time per station | <b>3-5 minutes</b>         |
| Total time       | <b>20-40 min + debrief</b> |

### 3.3 Formative Assessment (<2 Minutes, No Grading)

#### Hattie Effect Sizes

| Influence               | Effect Size     |
|-------------------------|-----------------|
| Feedback (task-focused) | <b>d = 0.92</b> |
| Effects of testing      | d = 0.63        |
| Formative evaluation    | d = 0.48        |
| Questioning             | d = 0.48        |

#### Technique Quick Reference

| Technique                     | Time      | Protocol                                                                                         |
|-------------------------------|-----------|--------------------------------------------------------------------------------------------------|
| <b>Fist-to-Five</b>           | 15 sec    | "Show me fist to five" → students raise 0-5 fingers → scan room                                  |
| <b>Thumbs Up/Down</b>         | 10 sec    | 👍 =understand, 👎 =confused, 🤔 =unsure                                                            |
| <b>Mini Whiteboards</b>       | 2-3 min   | Question → Write → "3,2,1 SHOW!" → scan patterns                                                 |
| <b>Exit Tickets</b>           | 3-5 min   | 1-3 questions at lesson end (3-2-1, one-minute summary)                                          |
| <b>Cold Call + No Opt Out</b> | 30-60 sec | Question → think time → call name → if "IDK" → get answer from peer → return to original student |

#### Fist-to-Five Scale

| Fingers  | Meaning (Understanding) | Meaning (Agreement)   |
|----------|-------------------------|-----------------------|
| 0 (fist) | Don't understand at all | Completely disagree   |
| 1        | Need help               | Barely agree          |
| 2        | Need more practice      | Have concerns         |
| 3        | Understand pretty well  | Agree with some parts |
| 4        | Mostly understand       | Mostly agree          |
| 5        | Could teach it          | Completely agree      |

#### Exit Ticket Formats

| Format             | Structure                                          |
|--------------------|----------------------------------------------------|
| 3-2-1              | 3 things learned, 2 interesting points, 1 question |
| One Question       | Single application question from objective         |
| Traffic Light      | Green/Yellow/Red sticker                           |
| One-Minute Summary | Brief summary of main concepts                     |

## MODULE 4: Evidence-Based Lecture Techniques

### 4.1 Interactiveness & Pacing

#### The 10-2 Rule

| Segment         | Duration              | Purpose             |
|-----------------|-----------------------|---------------------|
| Lecture chunk   | <b>10 minutes max</b> | Focused instruction |
| Processing time | <b>2 minutes min</b>  | Student interaction |

Research basis: Costa (1981), Long, Swain & Cummins (1996) (IEE)

#### Processing Activities ("2 Minutes")

- Say Something: pairs make predictions, ask questions, clarify confusion
- Quick-write: summarize key point in 2 sentences
- Turn-and-talk: explain concept to partner
- Concept check: mini whiteboard response

#### Attention Research (Wilson & Korn, 2007)

The "10-15 minute attention span" is **not well-supported** by research:

- Studies showed high variability (10-18 minute lapses)
- Note-taking declined but retention did NOT
- 2010 chemistry study: attention breaks at **30 seconds, 4-5 minutes, 7-9 minutes**—more frequent than theory suggests

**Key finding:** Active learning has "dual benefits"—engaging attention during a segment AND refreshing attention after.

#### Hattie Effect Sizes for Instruction



| Method              | Effect Size          |
|---------------------|----------------------|
| Teacher clarity     | <b>d = 0.75</b>      |
| Scaffolding         | d = 0.82             |
| Feedback            | d = 0.70             |
| Direct instruction  | <b>d = 0.59-0.60</b> |
| Questioning         | d = 0.48             |
| Reciprocal teaching | d = 0.74             |

**Direct Instruction (Hattie definition)** is NOT "sage on the stage" lecturing. It includes:

1. Clear learning intentions and success criteria
2. Worked examples and models
3. Guided practice with feedback
4. Formative checks for understanding
5. Lesson closure

### Interleaving Effect (Make It Stick)

| Context              | Effect Size     |
|----------------------|-----------------|
| Overall interleaving | <b>g = 0.42</b> |
| Visual materials     | g = 0.67        |
| Mathematical tasks   | g = 0.34        |
| Memory benefit       | up to g = 0.65  |
| Transfer benefit     | up to g = 0.66  |

*Warning: Interleaving feels harder (students prefer blocking), but produces better retention.*

## 4.2 Questioning Strategies

### Wait Time (Mary Budd Rowe, 1972)

| Measure                       | Baseline    | Optimal             |
|-------------------------------|-------------|---------------------|
| Wait Time I (after question)  | 0.7-1.4 sec | <b>3-5 seconds</b>  |
| Wait Time II (after response) | 0.9 sec     | <b>3-5 seconds</b>  |
| Critical threshold            | —           | <b>2.7+ seconds</b> |

### Effects of Increased Wait Time (3-5 sec)

### *Student Changes:*

- ↑ Response length and completeness
- ↑ Voluntary participation
- ↑ Student confidence
- ↑ Speculative/hypothesis responses
- ↑ Student-to-student interaction
- ↑ Higher-order thinking
- ↓ "I don't know" responses
- ↑ Participation from "slow" students

### *Teacher Changes:*

- Increased response flexibility
- More varied questioning
- Changed expectations for struggling students

### **Wait Time Protocol**

1. Ask question clearly
2. Count silently: "one Mississippi, two Mississippi, three Mississippi"
3. Avoid eye contact during wait (reduces pressure)
4. Select student
5. Student responds
6. Wait 3+ more seconds before reacting
7. "Anyone else with a response or comment?"

### **Cold Calling (Lemov - Teach Like a Champion)**

| Type          | Characteristics                     | Outcome                                                   |
|---------------|-------------------------------------|-----------------------------------------------------------|
| Collaborative | Warm, invitational, conversational  | Enhanced belonging, rich discussion, psychological safety |
| Unconditional | Confrontational, catching off-guard | Superficial engagement, creates anxiety                   |

### **Evidence-Based Cold Call Protocol**

**1. EXPLAIN THE WHY (first week):**

"This ensures everyone participates—it's not a 'gotcha'"

**2. GIVE THINK TIME (3-5 sec minimum):**

Ask question → Pause → Say name

**3. MAKE IT WARM, PREDICTABLE, UNIVERSAL:**

- Warm tone
- Positive expression
- Call on ALL students regularly

**4. USE COLLABORATIVE LANGUAGE:**

"Right, so Kelly, what were you thinking?"

"Who can build on what Marcus said?"

**5. MAINTAIN PSYCHOLOGICAL SAFETY:**

- Never use to "catch" inattentive students
- If struggling: "I'll ask others, then come back to you"
- Normalize errors

**No Opt Out Protocol (Lemov)**

IF student says "I don't know":

Format 1: Teacher provides answer → Student repeats

Format 2: Peer provides answer → Student repeats

Format 3: Teacher provides cue → Student uses it

Format 4: Peer provides cue → Student uses it

THEN return to original student for correct answer

ACKNOWLEDGE: "Thank you. Great that you worked through that."

**Stretch It Protocol (Lemov)**

After correct answer, extend with:

| Type                | Example                                   |
|---------------------|-------------------------------------------|
| Ask How or Why      | "How did you arrive at that?"             |
| Ask for Another Way | "What's a different method?"              |
| Ask for Better Word | "Can you use a more precise term?"        |
| Ask for Evidence    | "What in the text supports that?"         |
| Ask for Integration | "How does this connect to [other topic]?" |
| Ask for Transfer    | "Where else might you see this?"          |

Bloom's Taxonomy Question Stems

| Level         | Type   | Stems                           |
|---------------|--------|---------------------------------|
| 1. Remember   | Lower  | What, who, when, list, define   |
| 2. Understand | Lower  | Explain, describe, compare      |
| 3. Apply      | Lower  | How would you use, solve, apply |
| 4. Analyze    | Higher | Why, how does X relate to Y     |
| 5. Evaluate   | Higher | Which is better, defend, assess |
| 6. Create     | Higher | Design, create, propose         |

Build from lower → higher progressively; scaffold up the taxonomy.

Quick Reference: Effect Size Summary Table

| Intervention                       | Effect Size   | Source        | AI Priority |
|------------------------------------|---------------|---------------|-------------|
| Self-reported grades/metacognition | d = 1.44      | Hattie        | HIGH        |
| Jigsaw method                      | d = 1.20      | Hattie        | HIGH        |
| Multimedia principle               | d = 1.67      | Mayer         | HIGH        |
| Temporal contiguity                | d = 1.30      | Mayer         | HIGH        |
| Feedback (task-focused)            | d = 0.70-0.92 | Hattie        | HIGH        |
| Scaffolding                        | d = 0.82      | Hattie        | HIGH        |
| Teacher clarity                    | d = 0.75      | Hattie        | HIGH        |
| Reciprocal teaching                | d = 0.74      | Hattie        | HIGH        |
| Spaced practice                    | d = 0.71      | Hattie        | HIGH        |
| Metacognitive strategies           | d = 0.69      | Hattie        | HIGH        |
| Retrieval practice                 | g = 0.50-0.61 | Meta-analyses | HIGH        |
| Direct instruction                 | d = 0.59-0.60 | Hattie        | MEDIUM      |
| Flipped classroom                  | d = 0.58      | Hattie        | MEDIUM      |

| Intervention         | Effect Size | Source        | AI Priority |
|----------------------|-------------|---------------|-------------|
| Worked examples      | d = 0.57    | Hattie        | MEDIUM      |
| Practice testing     | d = 0.54    | Hattie        | MEDIUM      |
| Questioning          | d = 0.48    | Hattie        | MEDIUM      |
| Interleaving         | g = 0.42    | Meta-analysis | MEDIUM      |
| Cooperative learning | d = 0.40    | Hattie        | MEDIUM      |

**Hattie Hinge Point: d = 0.40** (average educational intervention effect)

- d > 0.40 = above average impact
- d = 0.50 ≈ one grade level gain
- d = 1.00 ≈ two grade levels

## Key Timing Specifications Reference

| Parameter                       | Value               |
|---------------------------------|---------------------|
| Lecture chunk maximum           | 10 minutes          |
| Processing time minimum         | 2 minutes           |
| Wait Time I (after question)    | 3-5 seconds         |
| Wait Time II (after response)   | 3-5 seconds         |
| Wait time threshold for benefit | 2.7+ seconds        |
| Pre-class video maximum         | 6-12 minutes        |
| Working memory duration         | ~20 seconds         |
| Working memory capacity         | 3-4 chunks          |
| Content segment for chunking    | 5-7 minutes         |
| Spaced review schedule          | Day 1 → 3 → 7 → 14+ |

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